

JioHotstar SQL Analysis Project

Dataset: JioHotstar

Database: MySQL

Project objective: Validate the dataset and extract business insights using SQL.

Executive Summary:

The dataset contains **6,874 records**, covering movies and TV content released from **1928 to 2023**.

The project demonstrates data validation, aggregation, filtering, subqueries, CTEs, and window functions.

Section A — Data Validation

Q.1 What is the total number of records in the dataset?

Result: 6,874 records.

Insight: The dataset contains 6,874 records available for analysis.

Q.2 Are there any NULL values in key columns (Title, Genre, Release_year, age_rating, Running_time, content_Type)? If yes, how many per column?

Result: All checked columns returned 0 NULL values.

Insight: The key fields are complete for analysis.

Q.3 Are there any duplicate titles in the dataset?

Result: Repeated title names exist; Rubaru Roshni appears 7 times, Guru 5 times, and several titles appear 3–4 times.

Insight: Repeated titles are not automatically duplicate records. Title alone should not be treated as a unique key because their genre are different.

Q.4 How many distinct content types exist, what are they, count of movie and TV?

Result: Movie: 4,568; TV: 2,306.

Insight: There are two content types. Movies are the larger part of the catalog.

Q.5 How many distinct genres exist in the dataset?

Result: 37

Insight: There are 37 distinct genres exist in the dataset.

Q.6 What is the minimum and maximum release year present in the data?

Result: 1928 to 2023.

Insight: The data are about 95 years.

Q.7 Are there any records with unrealistic or invalid running times (e.g., 0 or negative)?

Result: Records with runtime 0 are present, including TV titles.

Insight: A runtime of 0 for TV should be treated as missing/not-applicable rather than a normal runtime value.

Section B — Basic Business Analysis

Q.8 Which are the top 10 genres by number of titles?

Result: Drama 2,043; Comedy 791; Romance 642; Action 619; Reality 401; Thriller 352; Family 263; Animation 240; Documentary 207; Sport 180.

Insight: Drama has a clear lead over the other genres.

Q.9 How many titles exist for each age rating category?

Result:

	age_rating	Title
▶	A	338
	PG	1
	U	1251
	U/A 13+	2980
	U/A 16+	1235
	U/A 7+	1069

Insight: U/A 13+ is the largest rating category.

Q.10 Which release year has the highest number of titles?

Result: 2022 with 609 titles.

Insight: 2022 is the peak release year.

Q.11 What is the average running time of all movies?

```
SELECT AVG(Running_time) AS Running_time
FROM jiohotstar_details
WHERE content_type = "movie";

SELECT COUNT(*) AS suspicious_movies
FROM jiohotstar_details
WHERE content_Type = 'movie' AND Running_time < 30;
```

Result: 98.7467 minutes (approximately 98.75) calculated across all 4,568 movie entries.

Insight: While the overall average comes to ~98.75 minutes, closer inspection revealed that 868 movie entries (~19% of all movies) have running times under 30 minutes. Manually reviewing these records showed a mixed pattern — some are genuinely short-form content (e.g., short films and concert specials like "Cars Toon: Hiccups" and "Olivia Rodrigo: driving home 2 u"), while others are clearly full-length feature films with an incorrect runtime of 1 minute (e.g., "Dr. Cabbie", "A Dog Year"), and at least one case ("Obi-Wan Kenobi") appears to be a TV series misclassified as a movie. This indicates a data quality issue in the source dataset rather than a calculation error. The 98.75-minute average should therefore be treated as directionally correct but affected by these outliers — a cleaned dataset excluding likely-erroneous entries would give a more accurate feature-film average.

```
SELECT AVG(Running_time) AS avg_runtime_excluding_outliers
```

```
FROM jiohotstar_details
```

```
WHERE content_Type = 'movie' AND Running_time >= 30;
```

Result: 118.87 minutes (excluding entries with runtime under 30 minutes)

Insight: After excluding the 868 likely-erroneous entries (runtime < 30 min), the average movie runtime rises from 98.75 to 118.87 minutes — a difference of exactly 20 minutes. This confirms that the outlier entries were meaningfully skewing the original average downward, and 118.87 minutes (~1 hour 59 minutes) is a more accurate representation of typical feature-film length on the platform, which aligns closely with standard industry runtime expectations (100–150 minutes for feature films).

Q.12 What is the average running time of TV Shows (if applicable)?

Result: 0.0000.

Insight: This is not a meaningful TV runtime average because TV runtime values are stored as 0 in the current data.

Q.13 How many titles were released each year?

Result:

	Release_year	number_of_title		Release_year	number_of_title		Release_year	number_of_title
▶	1928	1		2000	42		2015	355
	1937	1		2001	53		2016	423
	1940	2		2002	56		2017	513
	1941	5		2003	61		2018	449
	1942	1		2004	65		2019	498
	1947	2		2005	124		2020	2019
	1950	1		2006	105		2021	561
	1951	1		2007	139		2022	609
	1953	2		2008	154		2023	133
	1955	2		2009	179			
	1956	2		2010	202			
	1957	3		2011	194			
	1958	1						
	1959	5						

Insight: Content releases show a strong upward trend over time, especially after 2005 — from just single digits per year in the early-to-mid 1900s, to over 100 titles/year by 2005, and crossing 500+ titles/year by 2016-2021. The steepest growth happened between 2013 and 2022, which aligns with the global rise of OTT/streaming platforms and increased content acquisition. 2022 recorded the highest number of releases (609 titles). However, 2023 shows a sharp drop to just 133 titles — this is likely because the dataset was collected mid-year or doesn't capture the full year's releases yet, rather than an actual decline in content production.

Section 3 — Intermediate SQL

Q.14 What are the 10 longest movies by running time?

Result:

	Title	Running_time
▶	The 2022 Rock & Roll Hall of Fame Induction Ce...	229
	Daana Veera Soora Karna	226
	Dubai	199
	Titanic	194
	Kerala Varma Pazhassiraja	190
	Lajja	186
	Mallannaa	183
	Nayak	182
	The King & The Commissioner	182
	23rd March 1931 Shaheed	182

Insight: The 2022 Rock & Roll Hall of Fame Induction Ceremony is longest movie at 229 minutes (approximately 3 hours and 49 minutes).

Q.15 What are the 10 shortest movies by running time?

Result:

	Title	Running_time
▶	Swapner Din	1
	The Sandlot: Heading Home	1
	Dr. Cabbie	1
	Olivia Rodrigo: driving home 2 u	1
	Cars Toon: Hiccups	1
	Jane	1
	Royal Bengal Rahashya	1
	Sadgamaya	1
	Obi-Wan Kenobi	1
	Prema Aparadhi	1

Insight: All of above shortest movie and their running time is also same which is 1 minutes.

Q.16 How many titles were released in the last 5 years?

Result: 2665

Insight: In last five years 2665 titles were released.

Q.17 Which genres have more than 100 titles?

Genre	Total_titles
Action	619
Animation	240
Romance	642
Thriller	352
Comedy	791
Drama	2043
Documentary	207
Kids	104
Animals & Nature	119
Horror	118
Family	263
Sport	180
Reality	401

Result:

Insight:. Drama is the largest genre with 2,043 titles

Q.18 What is the average running time by genre, for movies only?

Result:

Genre	AVG_RUNNING_TIME
Action	130.6869
Animation	63.3824
Romance	115.7672
Thriller	95.7955
Comedy	110.7700
Drama	89.1708
Documentary	59.4254
Historical	118.5313
Superhero	125.9318
Mystery	98.8214
Kids	90.3111
Genre	AVG_RUNNING_TIME
Horror	84.4706
Family	105.6169
Adventure	94.2000
Musical	99.1333
Sport	145.5000
Science	38.3500
Food	42.0000
Concert Film	102.7500
Mythology	128.9333
Travel	18.0000
Shorts	6.7500
Fantasy	109.5714
Teen	86.6364
Docudrama	51.0000
Talk Show	124.0000

Insight: The average running time of sport is larger than other genre with 145.50.

Q.19 Which genres appear only in Movies but not in TV Shows (or vice versa)?

Result: Movie-only genres: None (empty result)

TV-only genres: Reality, Awards, Lifestyle, Formula E, Football, Kabaddi (6 genres)

Insight: Unlike movies, TV Shows have several exclusive genres that don't appear in the Movies category — Reality, Awards, Lifestyle, Formula E, Football, and Kabaddi. This makes intuitive sense, as these are inherently episodic or event-based formats (sports coverage, award ceremonies, reality shows) that align naturally with a TV/series structure rather than a single feature-length film. Conversely, every genre found in Movies also appears somewhere in TV content, suggesting Movies don't have any genre that's completely unique to that format.

Q.20 How many movies have a running time greater than the overall average running time?

Result: 2959 movies have a running time greater than the overall average (98.75 minutes)..

Insight: 2,959 out of 4,568 movies (approximately 65%) have a running time above the raw average of 98.75 minutes (see Q11). This majority-above- average pattern is expected given the earlier finding that a subset of entries (868 movies) have unrealistically low running times due to data quality issues — these outliers pull the average down, meaning a larger share of "normal" movies naturally fall above it. If calculated against the cleaned average of 118.87 minutes instead, this count would likely be lower.

Q.21 What is the single most popular genre (highest count)?

Result: Drama — 2,043 titles.

Insight: Drama is the most represented genre.

Q.22 Which content type has more titles for each age rating?

Result:

	age_rating	content_Type	total_titles
▶	A	movie	178
	A	tv	160
	PG	tv	1
	U	movie	868
	U	tv	383
	U/A 13+	movie	1824
	U/A 13+	tv	1156
	U/A 16+	movie	977
	U/A 16+	tv	258
	U/A 7+	movie	721
	U/A 7+	tv	348

Insight: Movie has more titles than TV for each age rating.

Section 4 — Advanced SQL & Window Functions

Q.23 Rank all genres by total number of titles, from highest to lowest.

```
WITH genre_count AS (
    SELECT Genre, COUNT(*) AS total_titles
    FROM jiohotstar_details
    GROUP BY Genre
)
SELECT Genre, total_titles,
       RANK() OVER (ORDER BY total_titles DESC) AS genre_rank
FROM genre_count;
```

Result:

Genre	total_titles	genre_rank
Drama	2043	1
Comedy	791	2
Romance	642	3
Action	619	4
Reality	401	5
Thriller	352	6
Family	263	7
Animation	240	8
Documentary	207	9
Sport	180	10
Animals & N...	119	11
Horror	118	12
Kids	104	13
Crime	99	14
Mythology	81	15
Talk Show	73	16
Superhero	63	17
Standup Co...	51	18
Adventure	49	19
Biopic	47	20
Mystery	42	21

Historical	42	21
Science Fiction	41	23
Science	34	24
Teen	31	25
Awards	28	26
Lifestyle	24	27
Food	20	28
Concert Film	18	29
Musical	16	30
Fantasy	11	31
Shorts	10	32
Travel	7	33
Docudrama	5	34
Formula E	1	35
Football	1	35
Kabaddi	1	35

Insight: Demonstrates CTE + RANK(). Drama has more titles and Formula E, football, kabaddi has only one titles.

Q.24 Calculate the year-over-year change in number of titles released.

Result:

Release_year	titles_released	prev_year_titles	yoy_change
1928	1	NULL	NULL
1937	1	1	0
1940	2	1	1
1941	5	2	3
1942	1	5	-4
1947	2	1	1
1950	1	2	-1
1951	1	1	0
1953	2	1	1
1955	2	2	0
1956	2	2	0
1957	3	2	1
1958	1	3	-2
1959	6	1	5
1960	1	6	-5
1961	3	1	2
1962	3	3	0
1963	3	3	0
1964	7	3	4
1965	10	7	3
1966	5	10	-5
1967	7	5	2
1968	6	7	-1
1969	9	6	3

Release_year	titles_released	prev_year_titles	yoy_change
1970	7	9	-2
1971	11	7	4
1972	13	11	2
1973	15	13	2
1974	11	15	-4
1975	6	11	-5
1976	7	6	1
1977	18	7	11
1978	15	18	-3
1979	14	15	-1
1980	15	14	1
1981	21	15	6
1982	23	21	2
1983	19	23	-4
1984	17	19	-2
1985	34	17	17
1986	32	34	-2
1987	27	32	-5
1988	33	27	6
1989	27	33	-6
1990	46	27	19
1991	42	46	-4
1992	53	42	11
1993	36	53	-17
1994	37	36	1

Release_year	titles_released	prev_year_titles	yoy_change
2000	42	54	-12
2001	53	42	11
2002	56	53	3
2003	61	56	5
2004	65	61	4
2005	124	65	59
2006	105	124	-19
2007	139	105	34
2008	154	139	15
2009	179	154	25
2010	202	179	23
2011	194	202	-8
2012	178	194	-16
2013	213	178	35
2014	241	213	28
2015	355	241	114
2016	423	355	68
2017	513	423	90
2018	449	513	-64
2019	498	449	49
2020	415	498	-83
2021	561	415	146
2022	609	561	48
2023	133	609	-476

Insight: Demonstrates LAG() for trend analysis. Notably, 2023 shows a sharp decline of -476 titles compared to 2022 — consistent with the earlier finding (Q13) that the dataset likely doesn't capture a full year of 2023 releases.

Q.25 Find the top 3 longest movies within each genre.

Result:

Genre	Title	Running_time	Genre	Title	Running_time
Action	Kerala Varma Pazhassiraja	190	Docudrama	Robert D'Souza The Dog Feeder	20
Action	The King & The Commissioner	182	Documentary	The Zen Diaries of Garry Shandling Part - II	138
Action	Ungalukkaga Oruvan	178	Documentary	Jane Fonda In Five Acts	128
Adventure	The Chronicles Of Namia: Prince Caspian	149	Documentary	Summer of Soul (...or, When the Revolution Could Not Be Tele...	117
Adventure	Star Wars: The Rise Of Skywalker	142	Drama	Lajja	186
Adventure	The Chronicles Of Namia: The Lion, The Witch And The Wardrobe	140	Drama	Mallannaa	183
Animals & Nature	Earth Live	93	Drama	23rd March 1931 Shaheed	182
Animals & Nature	Into The Okavango	93	Family	Anbarivu	165
Animals & Nature	Hostile Planet: Ultimate Survivors	89	Family	Life Is Beautiful	161
Animation	Dory's Reef Cam	181	Family	Bro Daddy	159
Animation	Arendelle Castle Yule Log	180	Fantasy	Daana Veera Soora Karna	226
Animation	Arendelle Castle Yule Log: Cut Paper Edition	180	Fantasy	BrahmĀstra Part One: Shiva	168
Biopic	Srimadvirat Veerabrahendra Swami Charitra	180	Fantasy	Aladdin	130
Biopic	Bhaag Milkha Bhaag	178	Food	Spencer & Robert's Game Day Buffet	42
Biopic	MS Dhoni: The Untold Story	178	Historical	Titanic	194
Comedy	Nanban	180	Historical	Hamilton	160
Comedy	Vandanam	179	Historical	Sarsenapati Hambirrao	151
Comedy	No 20 Madras Mail	170	Horror	Chandramukhi	165
Concert Film	The 2022 Rock & Roll Hall of Fame Induction Ceremony	229	Horror	Suryavanshi	161
Concert Film	Elton John Live: Farewell from Dodger Stadium	153	Horror	Kalpana	152
Concert Film	BTS: Permission to Dance on Stage - LA	130	Kids	Adbhutha Dweep	150
Crime	3 Char Sau Bees	156	Kids	Athisayan	135
Crime	The Big Bull	153	Kids	Dhoni	135
Crime	Khiladi	151	Musical	The Sound Of Music	174
Docudrama	Mike Tyson: Undisputed Truth	82	Musical	West Side Story	156

Insight: Demonstrates partitioned ranking. Interestingly, the overall longest movie (The 2022 Rock & Roll Hall of Fame Induction Ceremony, 229 min) ranks #1 within the Concert Film genre specifically, confirming the Q14 finding.

Q.26 Calculate a running total of titles released, ordered by year.

Result:

Release_year	titles_released	running_total	Release_year	titles_released	running_total	Release_year	titles_released	running_total
1928	1	1	1973	15	130	1999	54	912
1937	1	2	1974	11	141	2000	42	954
1940	2	4	1975	6	147	2001	53	1007
1941	5	9	1976	7	154	2002	56	1063
1942	1	10	1977	18	172	2003	61	1124
1947	2	12	1978	15	187	2004	65	1189
1950	1	13	1979	14	201	2005	124	1313
1951	1	14	1980	15	216	2006	105	1418
1953	2	16	1981	21	237	2007	139	1557
1955	2	18	1982	23	260	2008	154	1711
1956	2	20	1983	19	279	2009	179	1890
1957	3	23	1984	17	296	2010	202	2092
1958	1	24	1985	34	330	2011	194	2286
1959	6	30	1986	32	362	2012	178	2464
1960	1	31	1987	27	389	2013	213	2677
1961	3	34	1988	33	422	2014	241	2918
1962	3	37	1989	27	449	2015	355	3273
1963	3	40	1990	46	495	2016	423	3696
1964	7	47	1991	42	537	2017	513	4209
1965	10	57	1992	53	590	2018	449	4658
1966	5	62	1993	36	626	2019	498	5156
1967	7	69	1994	37	663	2020	415	5571
1968	6	75	1995	38	701	2021	561	6132
1969	9	84	1996	44	745	2022	609	6741
1970	7	91	1997	59	804	2023	133	6874

Insight: Demonstrates a running total with a window function. Titles released is increasing by year.

Q.27 Calculate each genre's percentage share of total content.

Result:

	Genre	total_titles	pct_share
▶	Drama	2043	29.72
	Comedy	791	11.51
	Romance	642	9.34
	Action	619	9.00
	Reality	401	5.83
	Thriller	352	5.12
	Family	263	3.83
	Animation	240	3.49
	Documentary	207	3.01
	Sport	180	2.62
	Animals & N...	119	1.73
	Horror	118	1.72
	Kids	104	1.51
	Crime	99	1.44
	Mythology	81	1.18

Insight: Drama has 29.72 percentage and mythology has 1.18 percentage.

Q.28 For each release year, rank genres by popularity within that year specifically.

Result:.

Release_year	Genre	total_titles	genre	Release_year	Genre	total_titles	genre	Release_year	Genre	total_titles	genre
1928	Family	1	1	1970	Kids	1	2	2022	Biopic	6	15
1937	Animation	1	1	1971	Drama	5	1	2022	Awards	6	15
1940	Animation	2	1	1971	Comedy	2	2	2022	Mythology	5	19
1941	Animation	4	1	1971	Thriller	2	2	2022	Adventure	4	20
1941	1941 Drama	1	2	1971	Kids	1	4	2022	Science ...	4	20
1942	Animation	1	1	1971	Romance	1	4	2022	Superhero	4	20
1947	Family	1	1	1972	Drama	6	1	2022	Food	4	20
1947	Animation	1	1	1972	Comedy	2	2	2022	Science	2	24
1950	Animation	1	1	1972	Thriller	2	2	2022	Sport	2	24
1951	Animation	1	1	1972	Family	2	2	2022	Historical	1	26
1953	Animation	1	1	1972	Romance	1	5	2022	Musical	1	26
1953	Romance	1	1	1973	Drama	8	1	2022	Talk Show	1	26
1955	Drama	1	1	1973	Romance	2	2	2022	Animation	1	26
1955	Animation	1	1	1973	Action	1	3	2022	Teen	1	26
1956	Mythology	1	1	1973	Comedy	1	3	2022	Formula E	1	26
1956	Romance	1	1	1973	Animation	1	3	2022	Travel	1	26
1957	Drama	2	1	1973	Thriller	1	3	2023	Drama	59	1
1957	Family	1	2	1973	Mythology	1	3	2023	Family	22	2
1958	Comedy	1	1	1974	Drama	4	1	2023	Reality	17	3
1959	Drama	3	1	1974	Romance	3	2	2023	Romance	8	4
1959	Family	1	2	1974	Thriller	2	3	2023	Mythology	6	5
1959	Science ...	1	2	1974	Adventure	1	4	2023	Documen...	5	6
1959	Animation	1	2	1974	Action	1	4	2023	Comedy	3	7
1960	Documen...	1	1	1975	Drama	3	1	2023	Sport	3	7
1961	Animation	1	1	1975	Comedy	1	2				

Insight: Shows how genre popularity changes over time. In earlier decades (1920s-1970s), Family and Animation dominated yearly rankings, while by 2022-2023, Drama consistently ranks #1 each year, with newer categories like Awards and Reality also appearing in top rankings — reflecting the platform's expanding content diversity in recent years.

Key Business Insights

- 6,874 records are available for analysis.
- Movies (4,568) outnumber TV records (2,306).
- Drama is the largest genre with 2,043 titles.
- U/A 13+ is the largest age-rating category with 2,980 titles.
- 2022 has the highest number of releases with 609 titles.
- Average movie runtime is approximately 98.75 minutes.
- No NULL values were found in the checked columns.
- Repeated titles should not automatically be treated as duplicate records because title alone is not a reliable unique identifier.
- TV runtime values of 0 should be documented as a data-quality limitation.
- The project demonstrates SQL skills including aggregation, subqueries, CTEs, GROUP BY/HAVING, RANK, ROW_NUMBER, LAG and running totals.
- TV Shows have 6 exclusive genres not found in Movies (Reality, Awards, Lifestyle, Formula E, Football, Kabaddi), reflecting their episodic/event-based format.